

Module 12

Business Finance

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION A – CASE QUESTIONS (Total: 50 marks)

Answer 1(a)

SRG should use PESTEL analysis to analyse the external business environment in order to identify the opportunity and threat to SRG in the short and long term.

Politically, the government has imposed very stringent rules to restrict the numbers of customers and the dine-in time of restaurants during the pandemic period, which affects the revenue of the restaurants in the short term. Nevertheless, the government also provides subsidies to the restaurants, which provide additional funding to SRG and help to support the restaurant business in the short term. It is a threat to the catering business if the government further strengthens the regulations to limit the number of diners at each table and reduces subsidies available to the operations of restaurants in the long term. On the other hand, it will be an opportunity for SRG if any of the existing dine-in restrictions are loosen or removed.

Economically, due to the pandemic, tight social distancing measures and the increase in unemployment, people are less willing to eat outside or spend. This will have an adverse impact on SRG' revenue and profitability due to its high fixed costs base, such as rent and staff costs, in the short term. However, increasing consumer demand of take-away food is clearly an opportunity for the restaurant industry as well as SRG, which, despite its lower margin, will allow SRG to recover part of the loss of dine-in revenue. It is also a threat to the catering business in the longer term because restaurants may not be able to fully recover all of the fixed costs with the lower margins generated by the take-away business.

Socially, due to the prolonged impacts of the pandemic, customers may change their dining habits and prefer to buy more take-away, have lunch in the office or eat at home. There are fewer dine-in customers, which will affect the revenue of SRG in short term. Nevertheless, customers may become more used to ordering take-away from restaurants, which opens another source of revenue to SRG. This is an opportunity for the restaurants to further develop their take-away businesses to meet the social changes in the long term.

Technologically, the pandemic has accelerated the deployment and application of online platforms and mobile applications for ordering in the food / catering industry. SRG has to invest and upgrade its ordering system to adapt to technological changes in the industry in the long term.

In addition, SRG can also leverage the use of technology to save on manpower and operating costs to improve profitability in the long term. This requires initial set up costs in the short term but would mean increased productivity and profitability in the longer term.

Environmentally, the take-away food business has been challenged for creating additional environmental waste, and restaurants have been urged to reduce the use of plastic food containers in take-away orders. Environmental groups have also exerted strong pressure on restaurants to use recycle food containers, which could increase the operating costs of the take-away businesses in both the short and long terms. In the short term, this is a threat to catering businesses as it involves higher operating costs but it also creates long term benefits due to the businesses becoming more environmentally friendly, and could attract more customers and support from the shareholders.

Legally, restaurants have had to invest significant amounts of money to comply with new health and safety regulations and standards. Due to the pandemic, the government has imposed more health and safety regulations, such as air quality and catering hygiene standards, that increases the operating costs of SRG in the short term. Nevertheless, these new standards and regulations can also enhance the confidence of customers to dine in, which also helps to benefit the restaurants in the long term.

For business to be more resilient and sustainable, SRG should take prompt actions to develop a take-away business in its operations and to upgrade the online ordering system to survive in the long term.

Answer 1(b)

As chief financial officer ("CFO") and certified public accountant ("CPA"), we should consider the ethical issue of counting part-time employees as full-time for getting more government subsidies; i.e. whether the suggestion of the director is acceptable ethically and in accordance with the Code of Ethics of the Hong Kong Institute of Certified Public Accountants.

The proposal is not ethically acceptable because the director is proposing to lie to the government, which would infringe the integrity principle that requires honesty and straight forwardness in all professional and business relationships under the Code of Ethics.

Moreover, the proposal of the director also infringes the principle of professional behaviour, which requires compliance with relevant laws and regulations, and the avoidance of any conduct that a professional accountant is aware of or should be aware of that could discredit the profession under the Code of Ethics.

The director's proposal could be discovered by independent, external auditors during an audit or the government review, which would damage the reputation of SRG and may lead to a potential lawsuit against SRG.

Therefore, as a CFO and CPA, the proposal should be rejected because it is a requirement for a CPA to provide professional services with knowledge and due care to SRG based on technical and professional standards and relevant legislation. SRG should not classify part-time employees as full-time employees when it applies for government subsidy.

Answer 2

An effective change management strategy and program is needed for successful implementation of changes to the business. The board of directors could make changes to the business by using Lewin's change management model and unfreezing the existing traditional catering culture of SRG, analysing the current practice and recognising what is not working under the pandemic situation, determining what requires new approaches or changes for business sustainability. This is an opportunity for SRG to review and evaluate existing business operations and processes and to come up with a new business strategy and digitalise the order processes.

The board of directors could explain how SRG will adapt to the current pandemic situation and survive by diversifying its take-away business and to digitalising the ordering processes. This is the most challenging part because successful changes have to overcome different barriers internally, such as existing employees, and externally, such as customers.

Finally, the board of directors has to refreeze the changes to make changes for the long term by setting up separate divisions in the organisation to handle take-away business and digitalisation of business processes; to update business vision and mission statements to show the change of business strategy and to promote a changing culture in SRG to cater for the future.

Based on Kotter's model, the board of directors could take the following steps to make the changes more successful and effective in SRG:

The board of directors has to generate a sense of urgency to shareholders so that they will agree to implement the new business strategy. The board of directors should explain to the shareholders that the business of SRG is in a critical situation and that SRG has to implement a new business strategy to digitalise the ordering processes to reduce operating costs and to start a take-away business to increase revenue; otherwise there will be a reduction in dividend payments at the end of the year.

The board of directors should try to create a guiding coalition with directors and managers of SRG to form a project team to work and to lead the changes. Shareholders will be informed in a timely manner about the changing status by the project team. This project team will coordinate with different stakeholders to promote and make the changes.

The board of directors should create a new vision that provides a clear statement for SRG to diversify its catering business from a dine-in to take-away business and also to digitalise its ordering processing to cater for the needs of the take-away business and to reduce operating costs.

The board of directors also needs to communicate the vision of SRG clearly by using internal memos and posts to let the whole group know about the change of business strategy. SRG could also revamp the website to distribute information about the new take-away strategy through a video that could be viewed by all stakeholders to communicate and reinforce the change.

The board of director should also empower the project team with formal terms of references to formally announce the aim and function of the project team and provide sufficient authority and resources to implement the digitalisation of business processes and the new take-away businesses.

The board of directors may consider starting the take-away business by promoting it on SRG's website and in restaurants, which could attract existing customers to try their foods using the new take-away services and could improve the revenue of SRG immediately and create some short-term wins to show that the new business strategy is workable and acceptable by existing customers. This would also allow SRG to receive feedback from customers to further improve its take-away business. SRG can also create a mobile application to promote the new take-away business.

SRG should also set up milestones to consolidate all short-term gains and take-away business experience to further diversify and digitalise its business operations through continuous improvement to create new dishes for customers and to promote its businesses online via different social media platforms.

The board of directors should institutionalise all changes by further digitalising other existing operation systems, recruiting additional staff to handle take-away businesses and changing the culture of SRG to accept new changes in the menu and dishes to further diversify its catering businesses.

Answer 3

(All figures are in thousands except ratios)

Net Present Value and Internal Rate of Return

Year	0	1	2	3	4	5	NPV	IRR
Net cash flow	- 2,000.00	2,000.00/5 +300.00- 100.00= 600.00	780.00	1,014.00	1,318.20	1,713.66		
Discount rate	1.00	0.8	0.64	0.51	0.41	0.33		
Discount net cash flow	- 2,000.00	480.00	499.20	517.14	540.46	565.51	602.31	
Discount rate 36%	1.00	0.74	0.54	0.40	0.29	0.21		
Discount net cash flow	- 2,000.00	444.00	421.20	405.60	382.28	359.87	12.95	
Discount rate 37%	1.00	0.73	0.53	0.39	0.28	0.21		
Discount net cash flow	- 2,000.00	438.00	413.40	395.46	369.10	359.87	- 24.18	36.35%

By adopting the NPV and IRR methods, even though the NPV is positive, SRG will not accept the project as it does not meet shareholders' required return of 40% (the IRR is below the shareholders' required return of 40%).

Payback

Cash flow

Year 1	600.00
Year 2	780.00
Total	1,380.00
Initial cost	- 2,000.00
Difference	- 620.00

Year 3 cash flow 1,014.00

Using the payback method, the project will be rejected as it requires 2 years and 7.3 months to pay back, which exceeds SRG shareholders' requirement of 2 years.

Accounting rate of return	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Accounting profit	200.00	280.00	392.00	548.80	768.32	2,189.12
Average annual accounting profit	437.82					
Accounting rate of return	21.89%					

Similarly, the project will also be rejected if ARR method is adopted as the rate of return is below 25%.

Return on investment	
Total accounting profit	2,189.12
Total cost of investment	2,000.00
Additional incremental cash flows for 5 years	500.00
	2,500.00
ROI =	87.56%

If ROI method of calculation is used, SRG should accept the project because the ROI exceeds shareholders' required return. However, this method of calculation ignores the time value of money and will not be an appropriate method for project appraisal

Recommendation:

Based on the above evaluations, it is recommended not to pursue the project. Amongst the various appraisal methods, NPV/IRR results are more reliable and should be used for the current investment decision because both the payback method and ROI ignore the time value of money. In addition, ROI is based on accounting profit, which could be affected by a change in accounting policy such as a change of depreciation policy within the five-year period, which may distort the evaluation.

Answer 4

SRG could finance its existing business, new take-away business, and digitalisation of its ordering system by internal funds, external debts and equity. According to the pecking order theory, different types of financing have different costs and considerations. The theory suggests companies should use the lower cost of financing sources before considering equity that requires a higher cost. Therefore, SRG should have a hierarchy of financing sources, starting from the lowest cost of financing such as retained earnings, cash or financial slack followed by external debts and equity. Based on the concerns of existing shareholders who do not want to dilute their investment ownerships and dividend payments, SRG should not consider issuing more equity to finance the existing business and new business initiatives unless there is no other better alternative.

However, the pecking order theory does not consider the impact of the change in capital structure on financial flexibility, risk, net income, control of the entity, and timing to SRG, and other factors that are unique to a business. Hence, SRG should also use the FRICTO framework to examine and evaluate each financing alternative before making a financing decision.

Flexibility refers to choices that can be made among financing alternatives in SRG. SRG has limited flexibility to issue new equity because existing shareholders do not want to have any change in their investment ownership and dividend payments. SRG has more flexibility to obtain debt financing because it has no current debts. Additionally, it is relatively easier or more flexible to repay debts than to buy back shares.

Risk refers to the ability of SRG to pay loan interest and repay principal to lenders even in adverse circumstances. A stress test could be used to test the sensitivity of changes in interest rates to the profitability, interest coverage and liquidity of SRG. SRG should have high interest coverage due to low debt liability but both profitability and liquidity are deteriorating in pandemic situations, which increases the financing risks. Gearing ratio may not be a concern to SRG due to low debt liability at this stage. Therefore, the use of debt financing poses less financial risks to SRG.

Income refers to the impact of different financing alternatives on returns to shareholders which is one of the main concerns of SRG. The income impact would be measured by earnings per share or return on equity, both of which reflect the profitability of SRG. SRG should compare these ratios with competitors in the catering industry to measure its profitability performance. The profitability of SRG is deteriorating so SRG should pay attention to the impact of an increase in interest rate on its profitability and its effect on dividend payments to existing shareholders.

Control refers to how different financing alternatives affect the ownership control of shareholders in SRG. Since the shareholders of SRG are worried about the dilution of their investment ownership, equity financing may not be used in this case. Moreover, the costs involved in rights issues might be high and may not be a feasible option for SRG. If SRG uses debt financing, SRG has to ensure that debt covenants are not too restrictive to impede the future growth and dividend payments of SRG.

Timing focuses on the current economic and capital market environment that may affect the financing alternatives available to SRG. Due to the adverse impacts of the pandemic on the economy, economic recovery may take a long time, which in turn may reduce the interest of investors to invest with uncertain returns and risks, so rights issues may not be attractive to SRG. SRG should also pay attention to possible future movement of the interest rate that could increase the cost of debt. Since the interest rate is at a lower level, SRG can obtain debt financing at a fixed interest rate that would avoid the fluctuating of interest rate and the uncertainty that could affect the profitability of SRG in the longer term.

SRG should also consider its future business plan and development in order to select financing alternatives to match the term and duration of financing alternatives to the length of the project. If SRG is raising financing for small and short-term projects, such as digitalisation of the ordering process, then short term financing, such as short- or medium-term bank loans, may be more appropriate. SRG may be required to provide collateral to obtain external financing.

* * * END OF SECTION A * * *

SECTION B – ESSAY / SHORT QUESTIONS (Total: 50 marks)**Answer 5(a)**

Proposal 1:

Coupons per year = HK\$400 million $\times$ 0.05 $\times$ (1 - 0.165) = HK\$16.7 million

Redemption value = HK\$400 million

Post tax cost of debt = 7% $\times$ (1 - 0.165) = 5.845 %

HK\$ in million	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Issuance of debenture	400					
Coupon		-16.7	-16.7	-16.7	-16.7	-16.7
Redemption						-400
	400	-16.7	-16.7	-16.7	-16.7	-416.7
Discount rate (5.845%)	1	0.94478	0.8926	0.84331	0.79674	0.75275
	400	-15.78	-14.91	-14.08	-13.31	-313.67
Total NPV =						28.25

The NPV is HK\$28,250,000, or approximately HK\$28 million.

Proposal 2:

Coupons per year = HK\$400 million $\times$ 0.04 $\times$ (1 - 0.165) = HK\$13.36 million

Redemption value = HK\$42 $\times$ 10 million = HK\$420 million

Post tax cost of debt = 7% $\times$ (1 - 0.165) = 5.845%

HK\$ in million	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Issuance of convertible bond	400					
Interests		-13.36	-13.36	-13.36	-13.36	-13.36
Redemption						-420
	400	-13.36	-13.36	-13.36	-13.36	-433.36
Discount rate	1	0.94478	0.8926	0.84331	0.79674	0.75275
	400	-12.62	-11.93	-11.27	-10.64	-326.21
Total NPV =						27.33

The NPV is HK\$27,330,000, or approximately HK\$27 million.

So the Company shall choose Proposal 1 with a higher NPV.

Answer 5(b)

Besides considering the fair value, the Company shall also consider qualitative consideration for Proposal 2.

Issuing a convertible bond would potentially dilute the shareholding as well as the control of the Company. If the share price at maturity is higher than the conversion price, the holder of the convertible bond would exercise the conversion, which would trigger issuance of new shares of the Company. Issuing new shares will also impact the return of investment of the Company.

However, there are also advantages for the convertible bond. The Company could settle the bond by issuing new shares without actual cash outflow and hence could reduce the pressure on the liquidity of the Company. The Company normally could offer a lower coupon rate on the bond with the convertible feature.

Nevertheless, whether the investor would convert the bond to equity depends on the market share price on maturity, which is not controllable by the Company. This would create uncertainty for the Company in its liquidity planning and create extra liquidity risk or cash surplus if the Company fails to estimate its share price accurately.

Finally, the valuation of the equity element of the convertible bond is more complicated compared with debenture, as it includes assumption of the future share price and could be subject to many variable factors, which is also considered as a challenge for the issuer of a convertible bond.

Answer 5(c)

For issuing debenture

In Hong Kong, any public offers of corporate bond, i.e. debenture, is required to obtain approval from the Securities and Futures Commission ("SFC") and the offers must comply with the requirements of the Companies (Winding Up and Miscellaneous Provisions) Ordinance (Cap. 32) ("CWUMPO").

The 3rd Schedule to the CWUMPO sets out the details to be included in the prospectus. The registration of the prospectus with the Registrar of Companies is required to be authorised by the SFC.

For issuing convertible bonds

On top of the requirements that any public offers for corporate bond in Hong Kong must obtain the approval from the SFC and the offers must comply with the requirements of the CWUMPO, as well as the requirement of the registration of the prospectus, Chapter 28 of the Main Board Listing Rules has provided the respective rules for any issuance of convertible debt securities:

- (a) Before the issuance of all convertible debt securities, the Hong Kong Exchanges and Clearing Limited ("HKEx") should be consulted at the earliest opportunity and approval by the HKEx should be obtained before they can be issued.
- (b) The Company must comply with both the requirements applicable to the debt securities for which listing is sought and those applicable to the underlying equity securities to which such convertible debt securities relate.

Answer 6(a)

- (i) Without any hedging:
$$= \text{JPY}129,000,000 / 131 - \text{JPY}129,000,000 / 129$$
$$= - \text{EUR}15,267.18$$

Exchange loss is EUR15,267.18.
- (ii) Hedging the whole receivables balance by buying the 3-month call option strike at 132: Rainbow bought the option but would not exercise the option in this case.
$$= \text{JPY}129,000,000 / 129 \times 1.07\% + (\text{JPY}129,000,000 / 131 - \text{JPY}129,000,000 / 129)$$
$$= - \text{EUR}10,700 - \text{EUR}15,267.18$$
$$= - \text{EUR}25,967.18$$

Loss is EUR25,967.18, including the cost of option and the exchange loss.

Answer 6(b)

- (i) Without any hedging:
= JPY129,000,000 / 134 - JPY129,000,000 / 129
= - EUR37,313.43
Exchange loss is EUR37,313.43.
- (ii) Hedging the whole receivables balance by entering into the 3-month forward contract:
There will be a loss by entering into the forward contract of EUR 22,727.27 but Rainbow could prevent a potential loss of EUR37,313.43.
So the saving would be
EUR37,313.43 - EUR22,727.27 = EUR14,586.16
- (iii) Hedging the whole receivables balance by buying the 3-month call option strike at 132:
The cost of the option is JPY129,000,000 / 129 x 1.07% = - EUR10,700
The realised loss for exercising the option
= JPY129,000,000 / 132 - JPY129,000,000 / 129 = - EUR22,727.27

Rainbow will exercise the option in this case so Rainbow could prevent a potential loss of EUR37,313.43.

So, the savings would be

$$\text{EUR}37,313.43 - \text{EUR}22,727.27 - \text{EUR}10,700 = \text{EUR}3,886.16$$

Answer 6(c)

A forward contract is a contract that agrees on a rate today for an exchange to take place on a specified date in the future.

The advantage of using forward contracts is that they can provide exact hedges for exposure. This is relatively convenient if we are able to construct the contract with a counterparty such as a bank. Another advantage of a forward is that there is no need to pay an upfront premium. However, there are also some disadvantages. A forward contract is a private and customisable agreement that settles at the end of the agreement and is traded over-the-counter. The contract price is a negotiated price between the parties which lacks transparency, and the price is not readily available on the market. Quotes from different banks are obtained for benchmarking purposes. Unlike futures and options contracts, the agreement and their details are not made public. There is a relatively high counterparty risk as the other party may default.

Options prices are normally more expensive than forward contracts which is also considered as an advantage of using forward contracts.

The advantage of using options for hedging is that they protect the Company against unfavourable moves of currency and allow opportunities for favourable moves of currency as it gives the buyer the "right" to choose whether to exercise or not. The products could be customised in a wide range of strategy options for hedging and are traded over-the-counter. However, the option premium reflects option payoffs which are more expensive.

A forward contract could possibly be a hedging instrument which perfectly matches all risks associated with a hedged item. It prevents ineffectiveness resulting from changes in the time value of money.

Answer 7(a)

To: Teddy

From: Keith

Subject: Recommendation for the most feasible performance measurement mechanism

Dear Teddy,

Referring to your questions about implementing a performance measurement mechanism for your new e-commerce business, I would like to provide the following comments for your consideration:

Summary of our evaluation between Balanced Scorecard and Performance Pyramid

Balanced Scorecard focuses on measuring and reporting how well an organisation performs, especially the return to shareholders, and includes various performance measures to assess whether the financial objectives are being achieved. The measures include profitability, liquidity and cash flow, return on assets, return on investment, as well as residual income and economic value added.

Balanced Scorecard provides management with information covering both financial and non-financial measures in four key perspectives namely: Financial, Customers, Internal Process, and Learning and Growth.

Balanced Scorecard mechanism mainly focuses on performance measurements of the company/organisation as a whole. On the contrary, Performance Pyramid, although is similar to Balanced Scorecard and provides multi-dimensional performance measures, it is not limited to company level performance measures, and can support various key hierarchical bands, thereby allowing company performances to be measured at lower level of the company hierarchy such as business unit level, as well as individual departments and work teams, etc.

The e-commerce business is a new business unit of Blanche. Thus, it is very important for Blanche to understand and monitor business performance at the business unit level in order for Blanche to better plan its resources as well as its further investment strategy for this business unit.

Blanche is required to measure this new e-commerce business which is expected to be segregated from its existing business as a separate business unit. With this in mind, Performance Pyramid is more suitable than Balanced Scorecard because it is able to provide multi-dimensional performance measurement at the business unit level, which would help Blanche more easily understand and keep track of its performance as well as the returns on its investment in the online infrastructure and IP platform.

Two suggested financial measures for Blanche

We suggest two financial measures for Blanche to measure the performance of the e-commerce business:

- (1) Return on investment (or assets) of the e-commerce business unit. Since there was a significant amount spent by Blanche to build the IT infrastructure as well as the online platform for this e-commerce business, it is important to understand whether the capital expenditure is able to generate satisfactory returns.
- (2) Divisional profitability of the e-commerce business unit. As profitability is an important indicator, a segregated profitability reporting for the e-commerce business unit is very important for Blanche to review the performance of this newly developed business.

I hope the above is helpful for you. If necessary, we can set up a call to further discuss this matter.

Best regards,

Keith

Answer 7(b)

Advantages of setting up a new legal entity for e-commerce business:

- (1) Easier to identify the performance of the e-commerce business. Blanche can directly review the performance from the financial statements of the new entity. The divisional financial information requires strong enterprise resource planning ("ERP") that Blanche may require to invest more in the ERP system.
- (2) Better accountabilities for the e-commerce business team. The e-commerce team is required to be responsible for the financial performance of the new company which provides better discipline on spending and liquidity management. It will lead to better balance sheet management.
- (3) Legal as well as commercial liabilities would be limited to this entity and would not affect the existing entity.
- (4) It would be convenient for any disposal or second listing in future.

Disadvantage of setting up a new legal entity for e-commerce business

- (1) This is more costly, because Blanche is required to pay extra professional fees such as audit fee and business registration fee.

- (2) It may require to develop another team of back-office function support such as IT, accounting, human resources, legal, as well as bank accounts management which may not be able to enjoy the synergy and economy of scales with the existing entity.
- (3) The recharge and intercompany financing between the existing entity and new entity creates extra administrative work.

Criteria to be considered for setting up a new legal entity for e-commerce business

- (1) Whether e-commerce business is large enough to justify being a separate legal entity.
- (2) Whether e-commerce business is part of the long-term strategic objective for Blanche or Blanche would like to "test the water" before committing to any structural business changes.
- (3) Whether the potential liabilities of the e-commerce business are significant and difficult to mitigate, so setting up a separate legal entity could help limit those risks.

Answer 7(c)

Social media harvesting is one of the methods to gather data from social media platforms automatically without drawing the attention of the users. Social media harvesting is different from traditional data gathering methodologies such as customer questionnaires and interviews. Blanche could develop some programs to integrate with various social medias, then it can extract targeted key words as well as the messages from social media platforms, such as the name or category of the products, to understand customer preference as well as developing targeted promotion plans.

The advantage is that it will not attract attention, so Blanche can easily get the natural reactions and preference from the customers, and the result would be more objective and reliable than that obtained from questionnaires or interviews.

Harvesting social media is programmed and would run in the background automatically. Once it is developed, Blanche will be able to keep accumulating the data for its study. In the long run this cost would be lower, compared with traditional ways of gathering data.

Finally, the data is real-time, and the volume of data would be huge compared with the traditional ways of data gathering and could provide immediate analysis. It shortens the turnaround time on data gathering and processing so it could allow Blanche to respond to any changes of preference or customer behaviours in a timely manner and could capture the latest market trend quicker.

* * * END OF EXAMINATION PAPER * * *